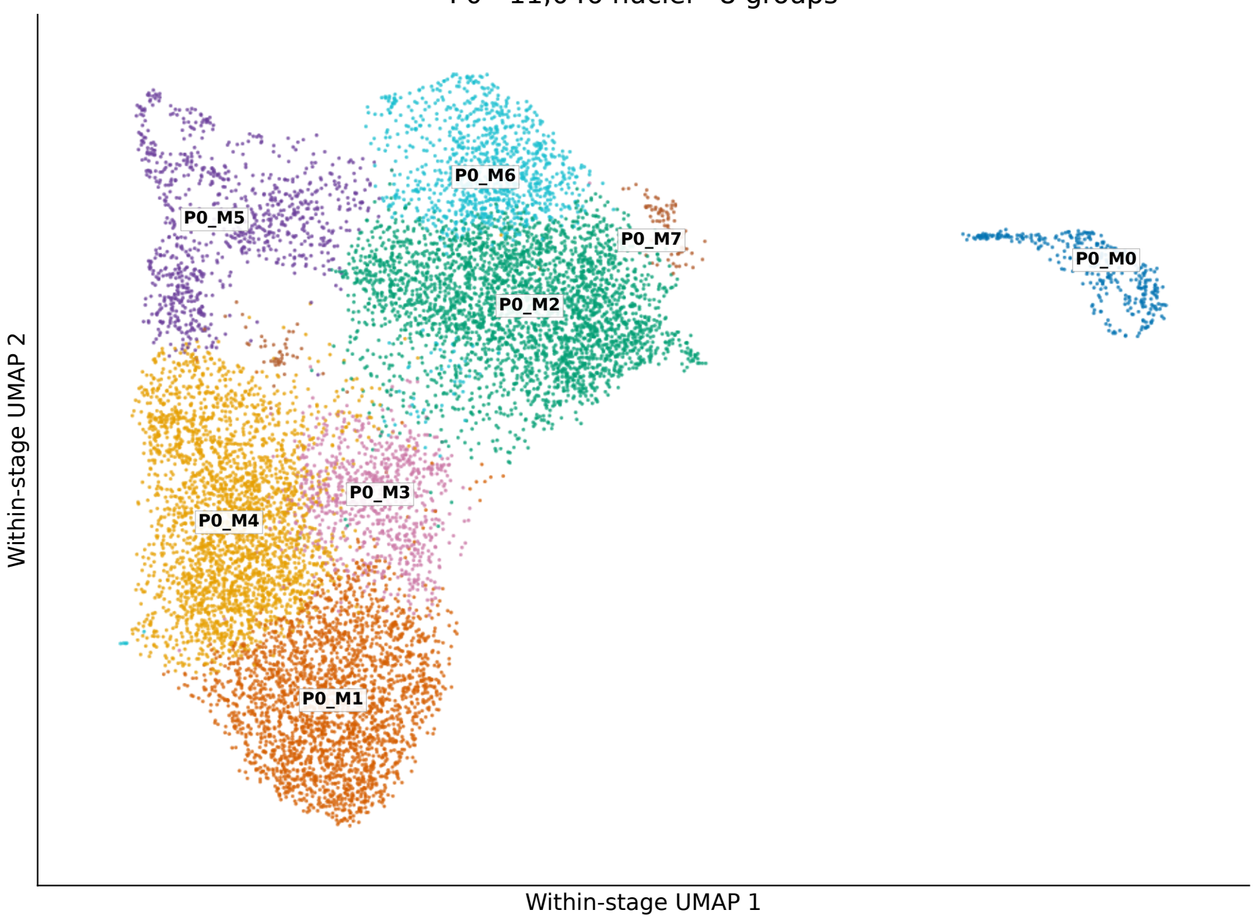
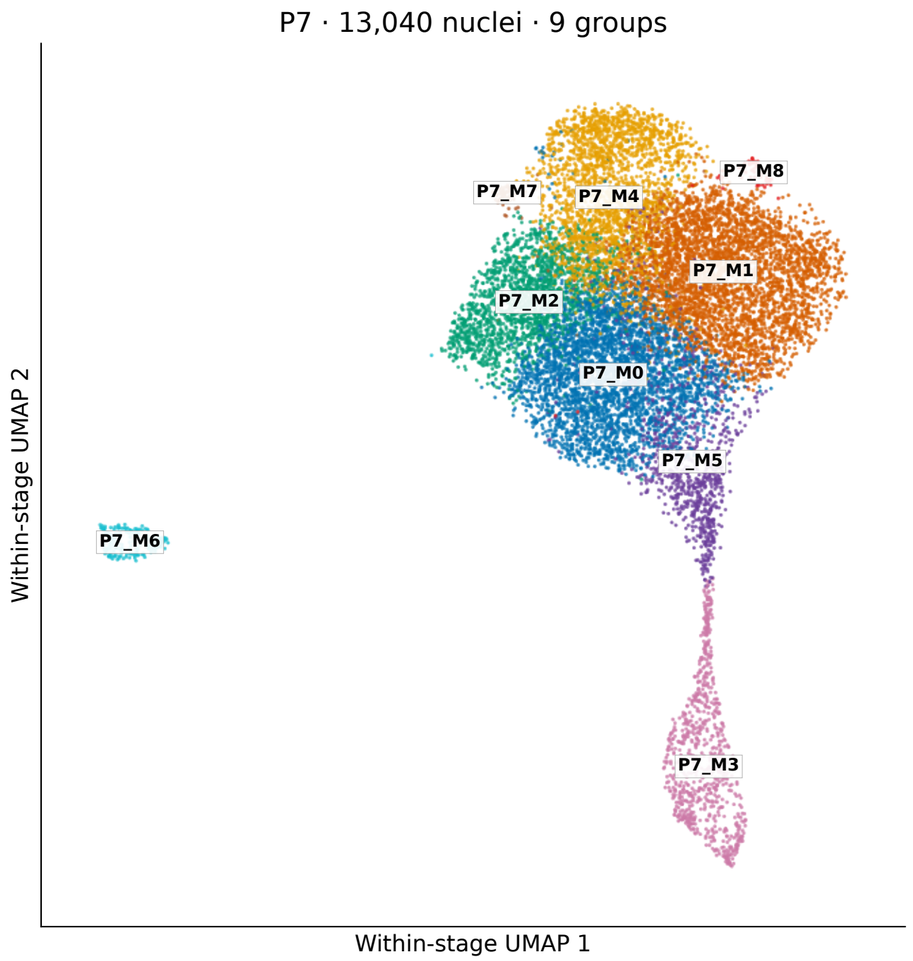
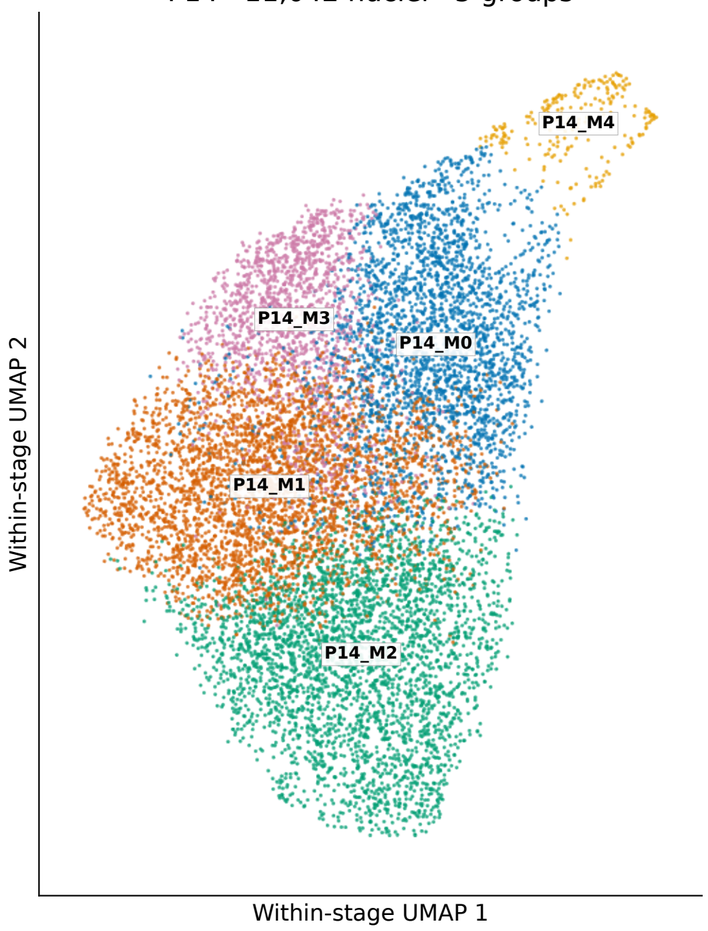
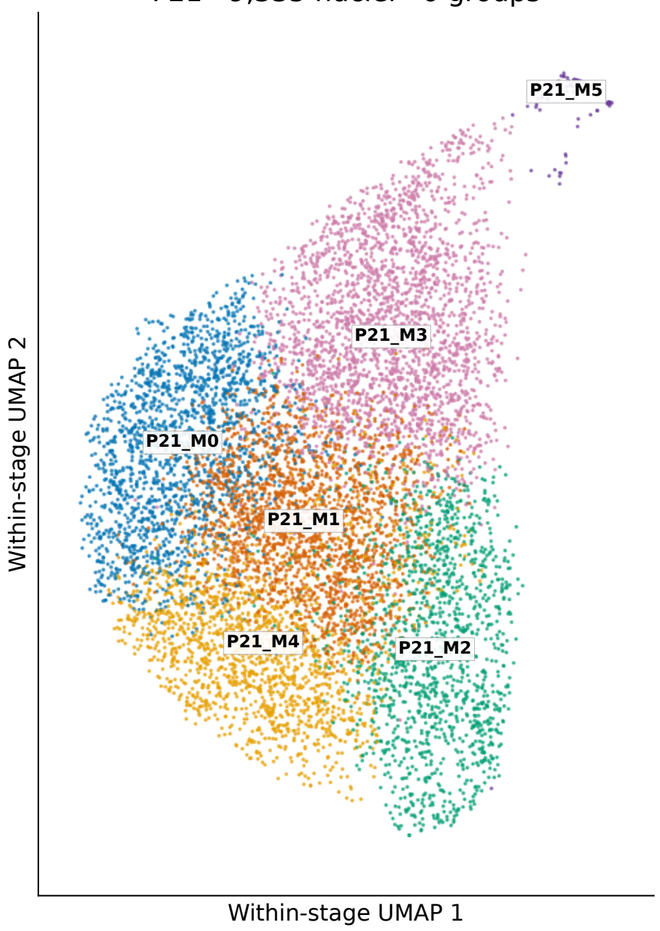
# Ventricular microenvironment-associated candidates within each stage



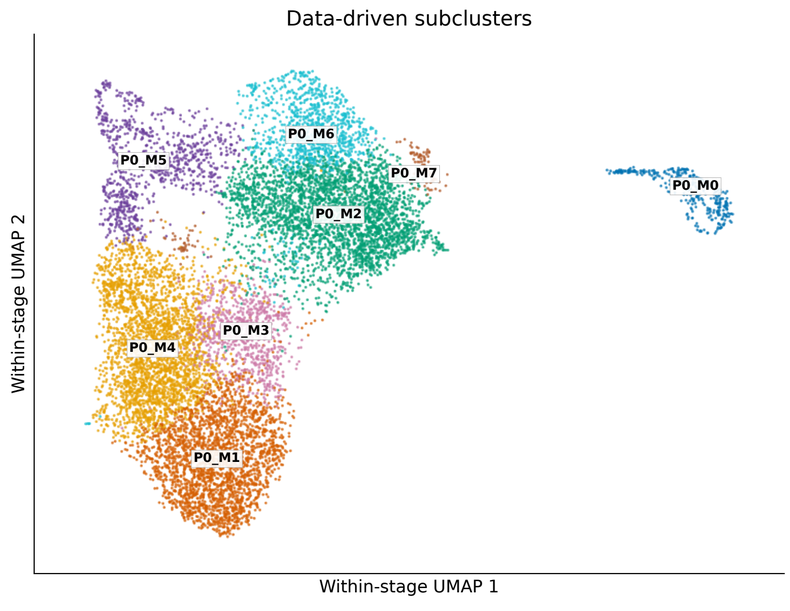



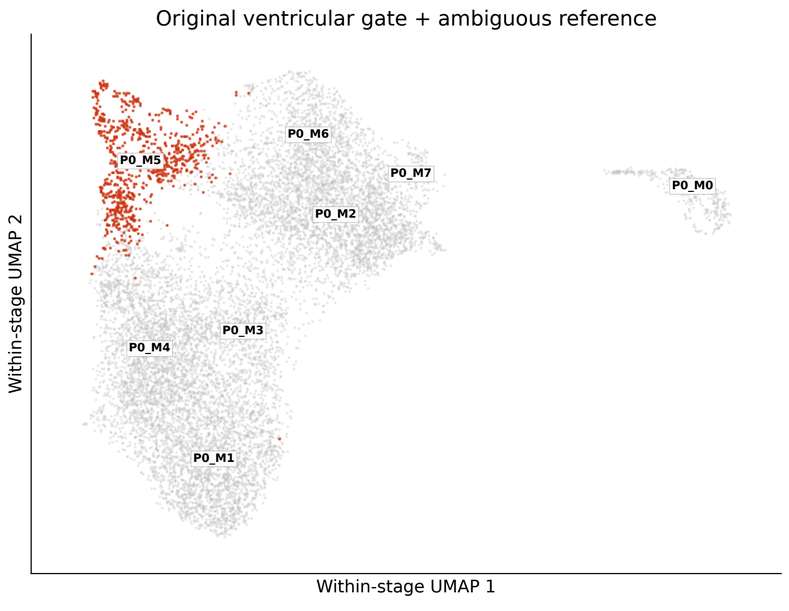

Independent coordinate spaces between stages · Colors/IDs identify computational groups; see stage pages for labels and evidence status
Candidate spatial or environmental interpretations require validation; the reference gate includes ambiguous Slit2-overlap cells.

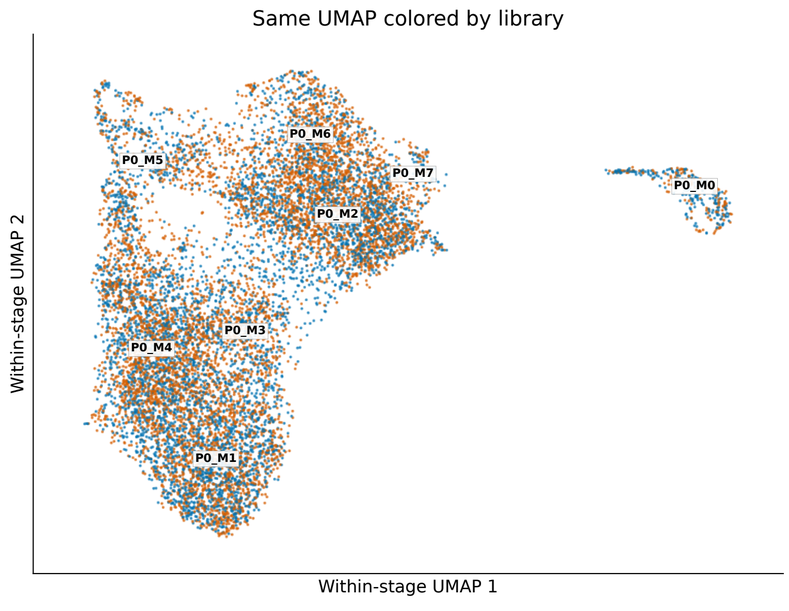
# P0: ventricular microenvironment-associated state candidates · n=11,046

core ventricular (n=10,315)
Slit2 overlap reference (n=731)

GSM8827071_P0_1 (n=5,091)
GSM8827072_P0_2 (n=5,955)



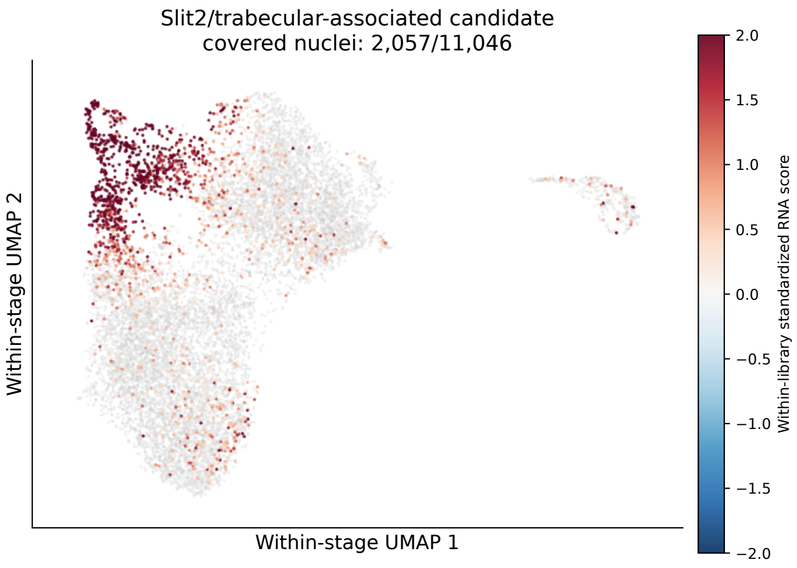



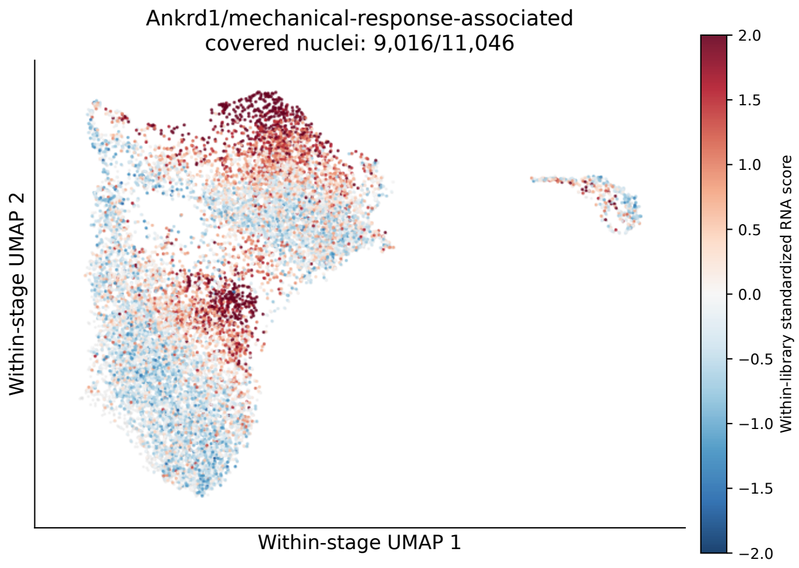



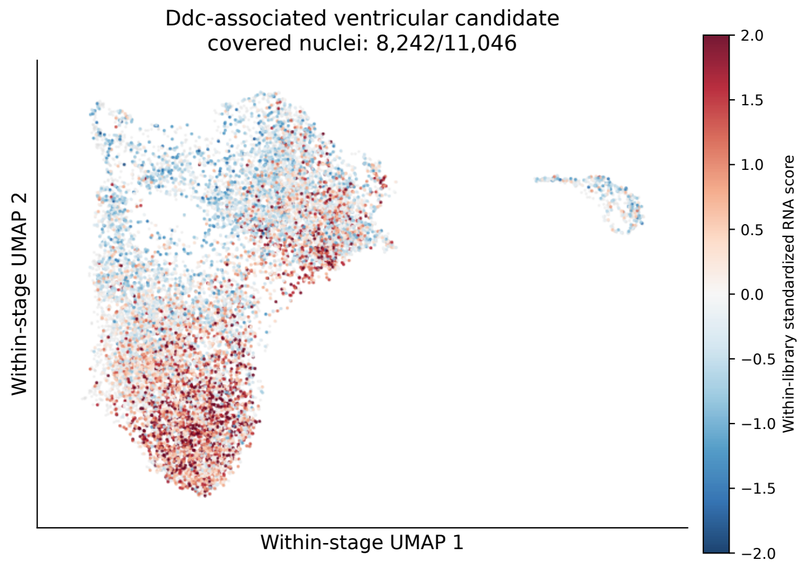

| Subcluster | Nuclei | Hypothesis-aligned label | Evidence / sensitivity status | Reference nuclei |
|---|---|---|---|---|
| P0_M0 | 325 | Cycling-associated control state | supported RNA state | 0 |
| P0_M1 | 2,393 | Ddc-associated ventricular candidate | supported RNA state | 1 |
| P0_M2 | 2,985 | Marker-defined: Zbtb16/Ncam1 | supported RNA state | 1 |
| P0_M3 | 796 | Ankrd1/mechanical-response-associated | candidate/review | 0 |
| P0_M4 | 2,588 | Fatty-acid oxidation genes | candidate/review | 19 |
| P0_M5 | 856 | Slit2/trabecular-associated candidate | supported RNA state | 706 |
| P0_M6 | 983 | Ankrd1/mechanical-response-associated | candidate/review | 3 |
| P0_M7 | 120 | Marker-defined: Pola1/Rad51b | candidate/review | 1 |

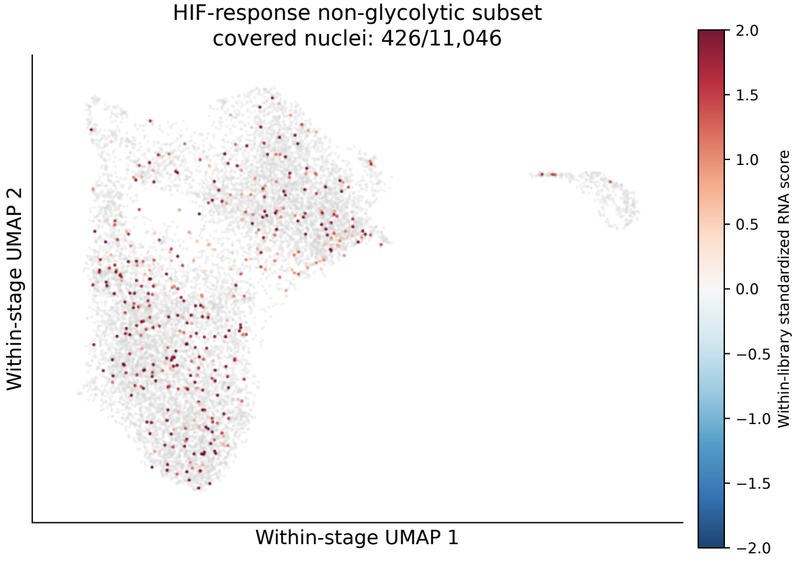
Same stage coordinates in all three panels · Ambiguous Slit2-overlap cells are a reference, not verified conduction or Purkinje cells
Labels are RNA-based hypotheses, not measured wall location, oxygen exposure or mechanical force. Other ages have independent coordinates.

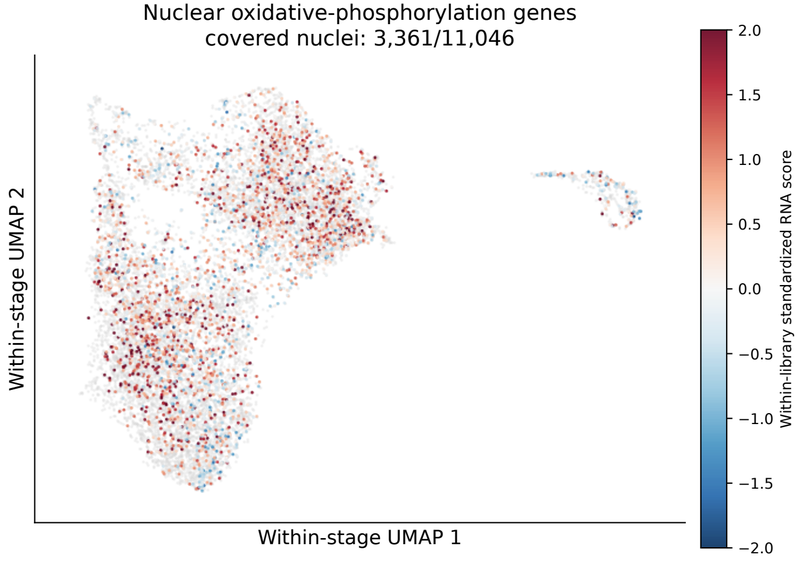
# P0: candidate spatial, oxygen-response and mechanical programs

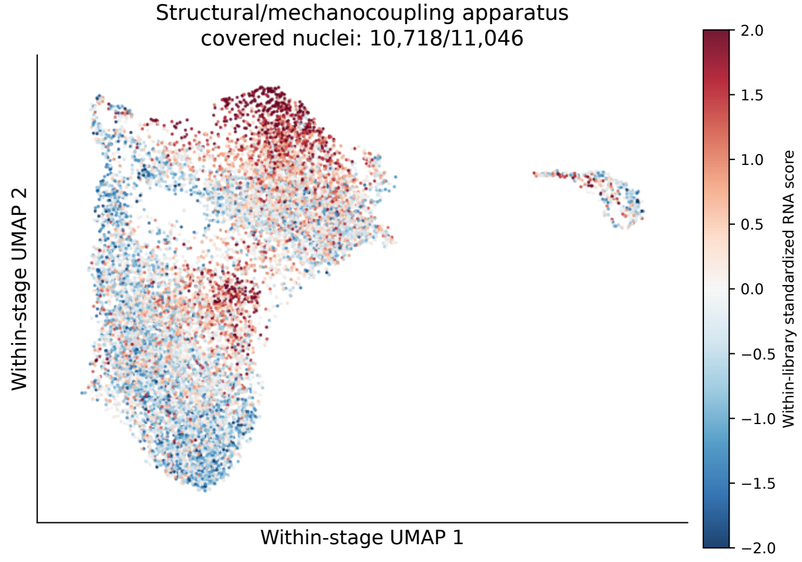


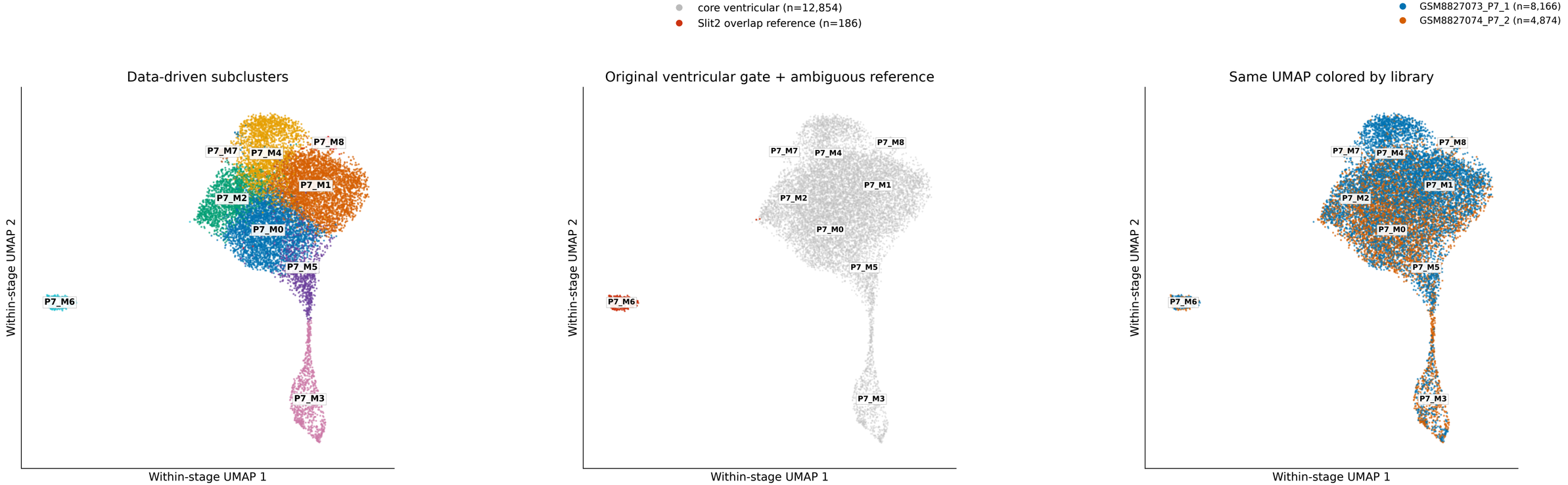




Same UMAP and axis limits throughout · Fixed color limits −2 to +2; scores are relative within libraries, not comparable oxygen measurements
Gray: nonfinite score or detected program-gene coverage <20% where coverage is supplied. Programs can overlap and form gradients.
RNA scores do not verify wall position, hypoxia, local force, or a discrete biological subtype.

# P7: ventricular microenvironment-associated state candidates · n=13,040

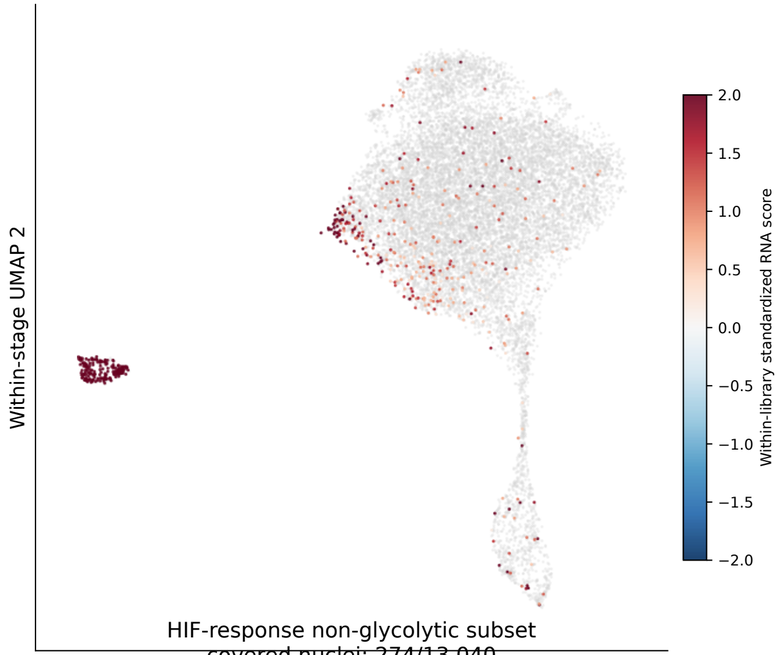



| Subcluster | Nuclei | Hypothesis-aligned label | Evidence / sensitivity status | Reference nuclei |
|---|---|---|---|---|
| P7_M0 | 3,357 | Marker-defined: Rreb1/Ppara | replicate-supported; sensitivity-dependent | 0 |
| P7_M1 | 3,676 | Ankrd1/mechanical-response-associated | supported RNA state | 0 |
| P7_M2 | 1,744 | Marker-defined: Fgf14/Pdzrn4 | candidate/review | 2 |
| P7_M3 | 776 | Cycling-associated control state | supported RNA state | 0 |
| P7_M4 | 2,293 | Marker-defined: Ivns1abp/Myom2 | candidate/review | 0 |
| P7_M5 | 888 | Marker-defined: Rad51b/Pola1 | replicate-supported; sensitivity-dependent | 0 |
| P7_M6 | 185 | Slit2/trabecular-associated candidate | supported RNA state | 184 |
| P7_M7 | 63 | Uncharacterized ventricular-enriched group | candidate/review | 0 |
| P7_M8 | 58 | Uncharacterized ventricular-enriched group | candidate/review | 0 |

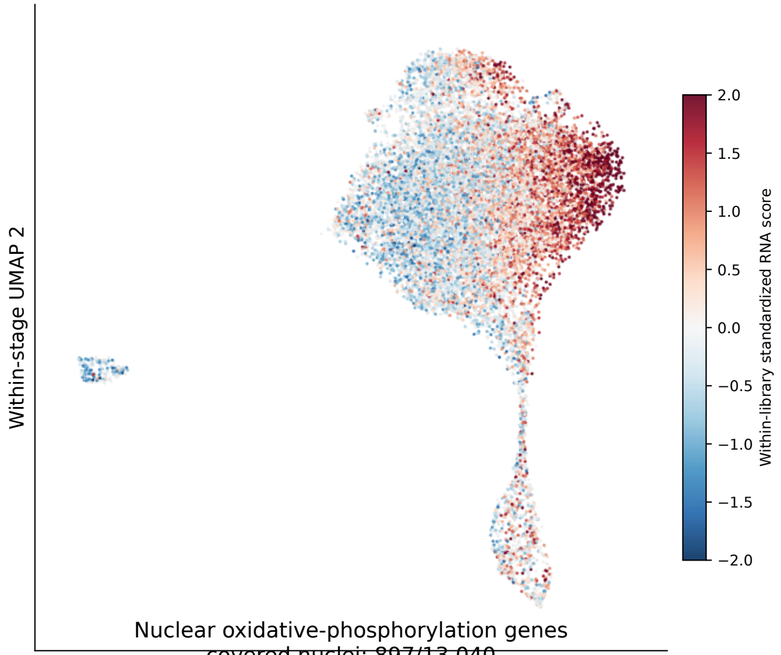
Same stage coordinates in all three panels · Ambiguous Slit2-overlap cells are a reference, not verified conduction or Purkinje cells
Labels are RNA-based hypotheses, not measured wall location, oxygen exposure or mechanical force. Other ages have independent coordinates.

# P7: candidate spatial, oxygen-response and mechanical programs

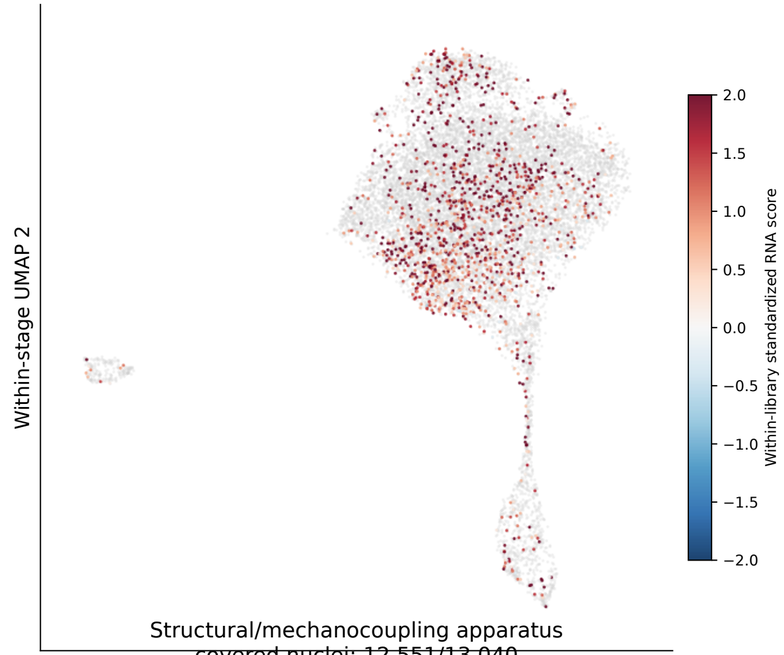
Slit2/trabecular-associated candidate
covered nuclei: 516/13,040



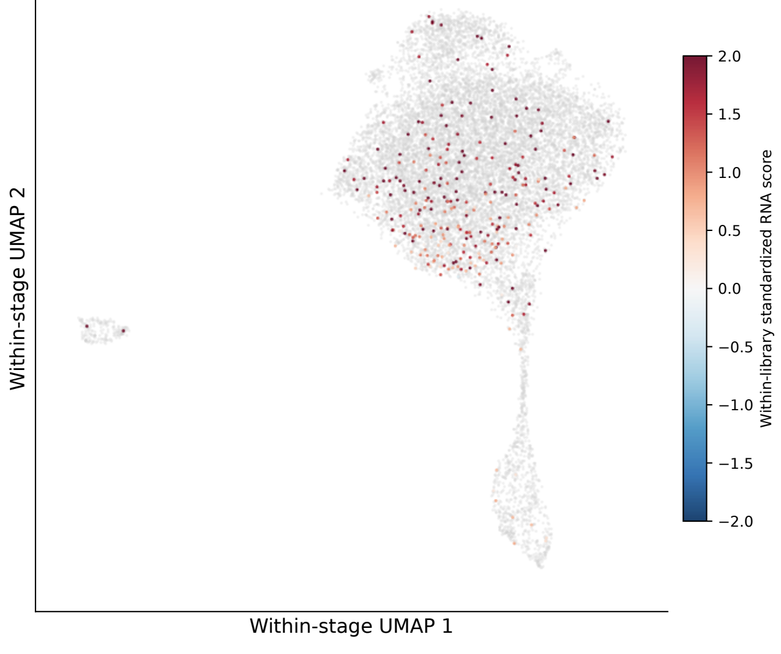

Ankrd1/mechanical-response-associated
covered nuclei: 10,798/13,040



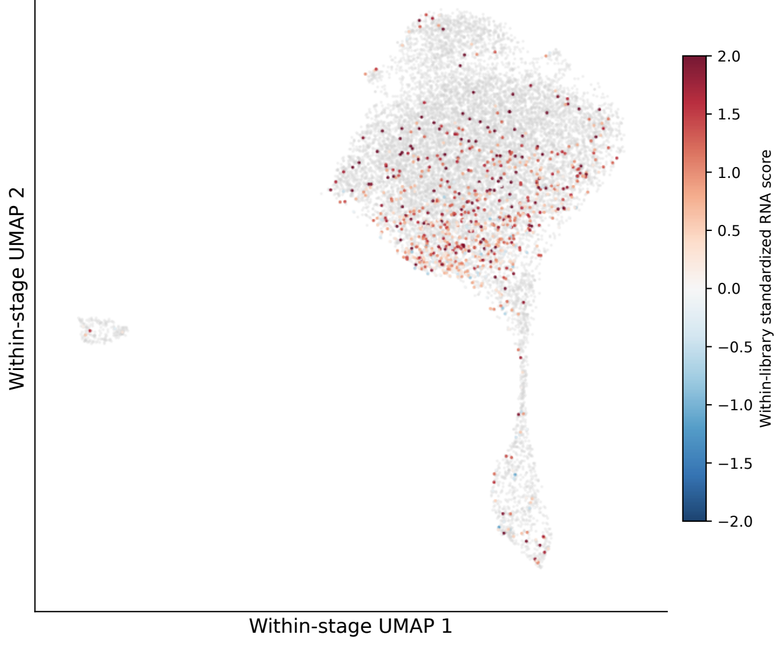

Ddc-associated ventricular candidate
covered nuclei: 1,302/13,040



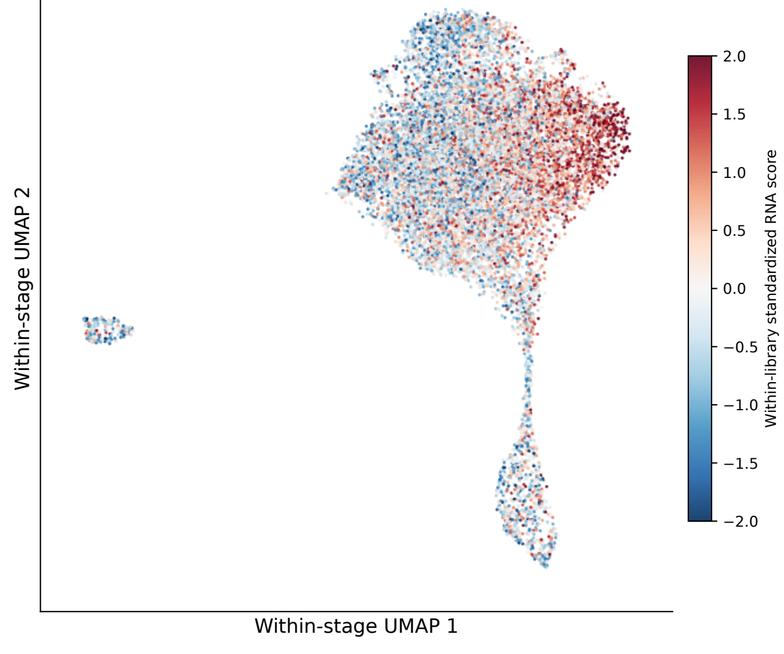

HIF-response non-glycolytic subset
covered nuclei: 274/13,040




Nuclear oxidative-phosphorylation genes
covered nuclei: 897/13,040

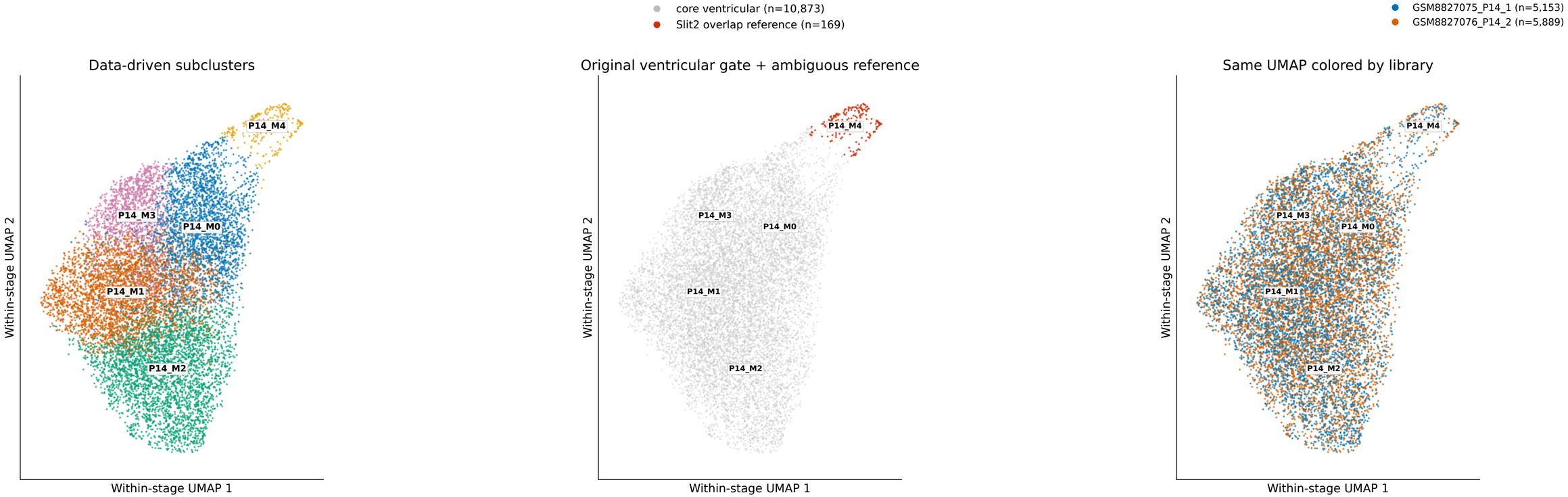



Structural/mechanocoupling apparatus
covered nuclei: 12,551/13,040




Same UMAP and axis limits throughout · Fixed color limits −2 to +2; scores are relative within libraries, not comparable oxygen measurements
Gray: nonfinite score or detected program-gene coverage <20% where coverage is supplied. Programs can overlap and form gradients.
RNA scores do not verify wall position, hypoxia, local force, or a discrete biological subtype.

# P14: ventricular microenvironment-associated state candidates · n=11,042

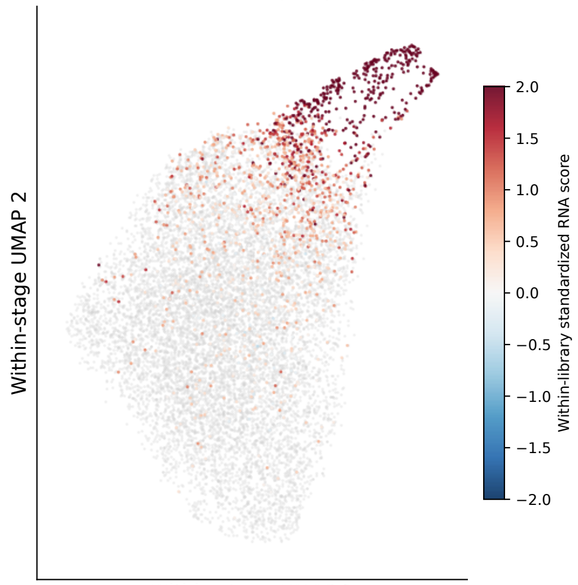



| Subcluster | Nuclei | Hypothesis-aligned label | Evidence / sensitivity status | Reference nuclei |
|---|---|---|---|---|
| P14_M0 | 2,638 | Slit2/trabecular-associated candidate | candidate/review | 0 |
| P14_M1 | 3,440 | Marker-defined: Lrtm1 | candidate/review | 0 |
| P14_M2 | 3,244 | Ankrd1/mechanical-response-associated | supported RNA state | 0 |
| P14_M3 | 1,501 | Marker-defined: Kcnd2/Pdzrn4 | candidate/review | 0 |
| P14_M4 | 219 | Slit2/trabecular-associated candidate | replicate-supported; sensitivity-dependent | 169 |

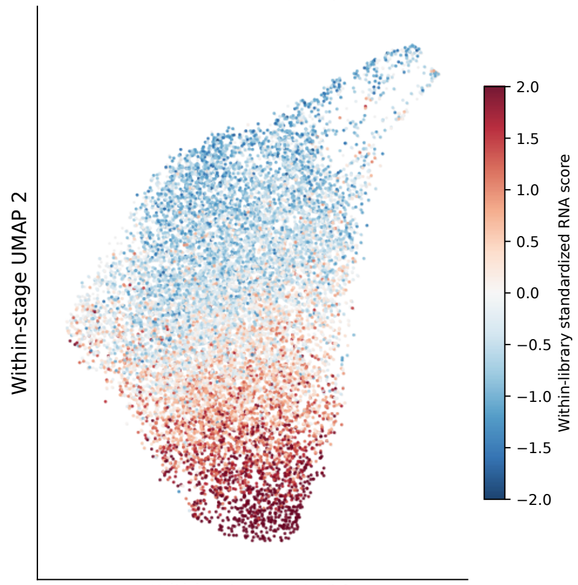
Same stage coordinates in all three panels · Ambiguous Slit2-overlap cells are a reference, not verified conduction or Purkinje cells
Labels are RNA-based hypotheses, not measured wall location, oxygen exposure or mechanical force. Other ages have independent coordinates.

# P14: candidate spatial, oxygen-response and mechanical programs

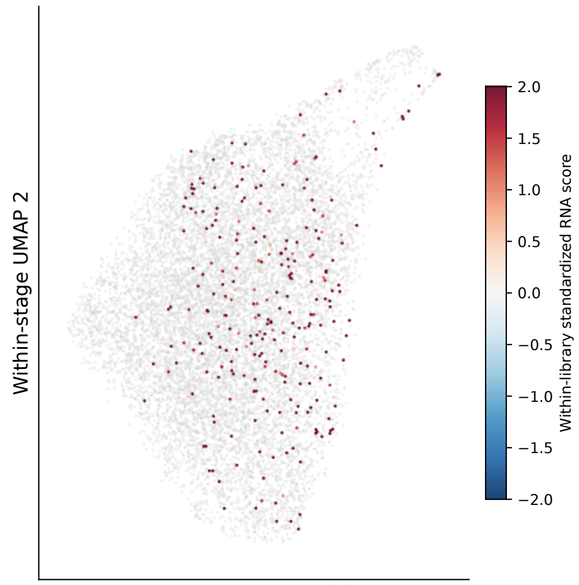
Slit2/trabecular-associated candidate
covered nuclei: 1,496/11,042



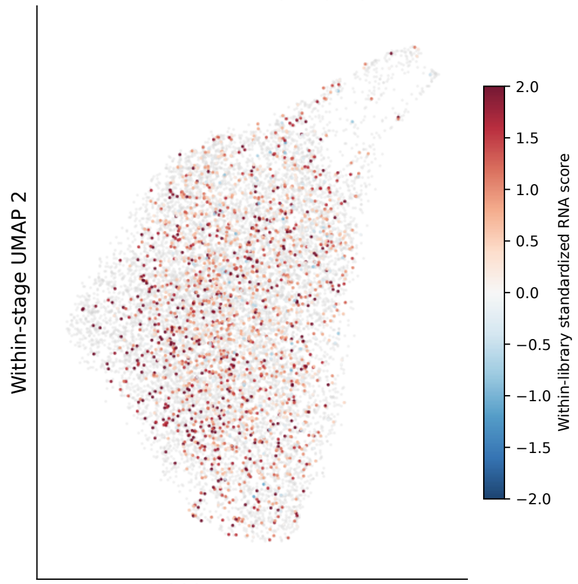

Ankrd1/mechanical-response-associated
covered nuclei: 10,446/11,042



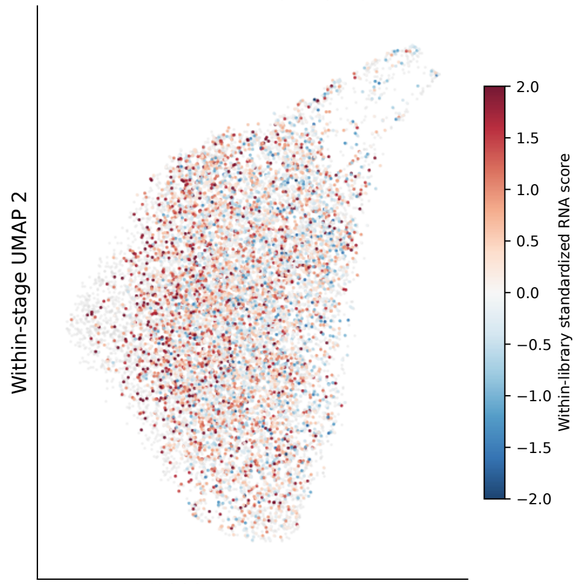

Ddc-associated ventricular candidate
covered nuclei: 262/11,042



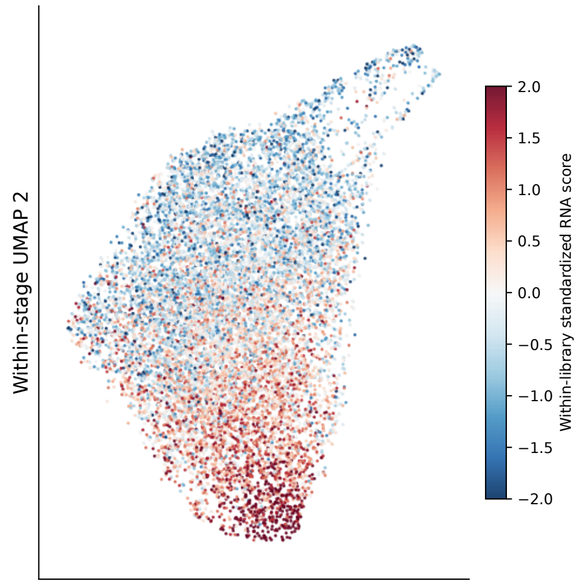

HIF-response non-glycolytic subset
covered nuclei: 2,088/11,042




Nuclear oxidative-phosphorylation genes
covered nuclei: 6,131/11,042

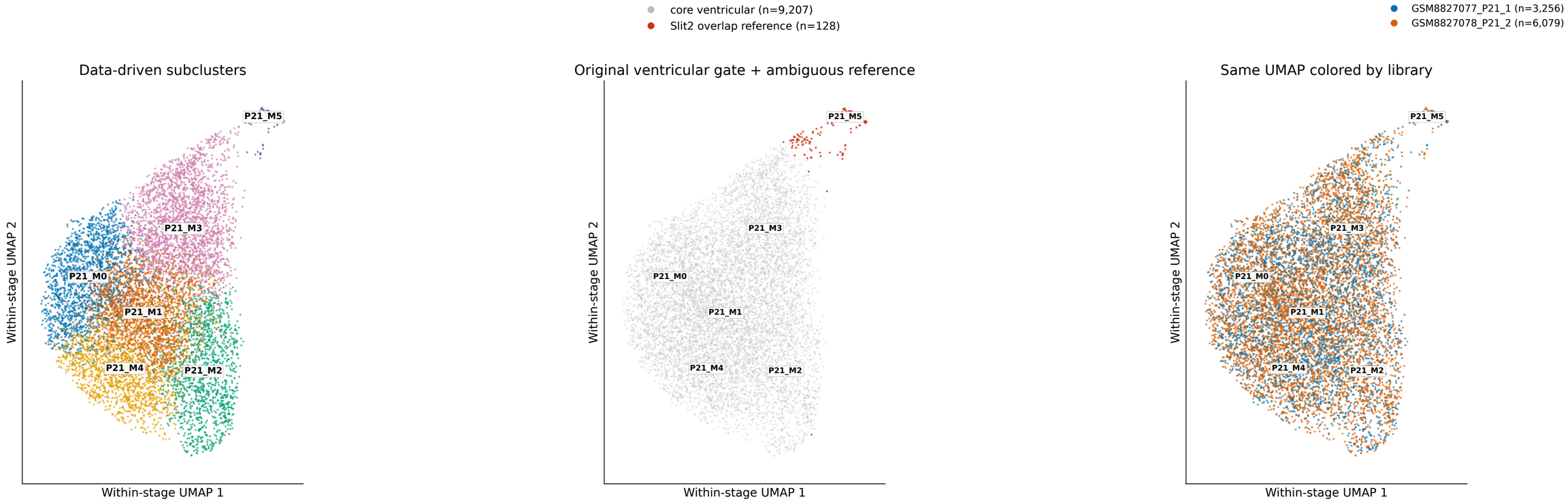



Structural/mechanocoupling apparatus
covered nuclei: 10,996/11,042




Same UMAP and axis limits throughout · Fixed color limits −2 to +2; scores are relative within libraries, not comparable oxygen measurements
Gray: nonfinite score or detected program-gene coverage <20% where coverage is supplied. Programs can overlap and form gradients.
RNA scores do not verify wall position, hypoxia, local force, or a discrete biological subtype.

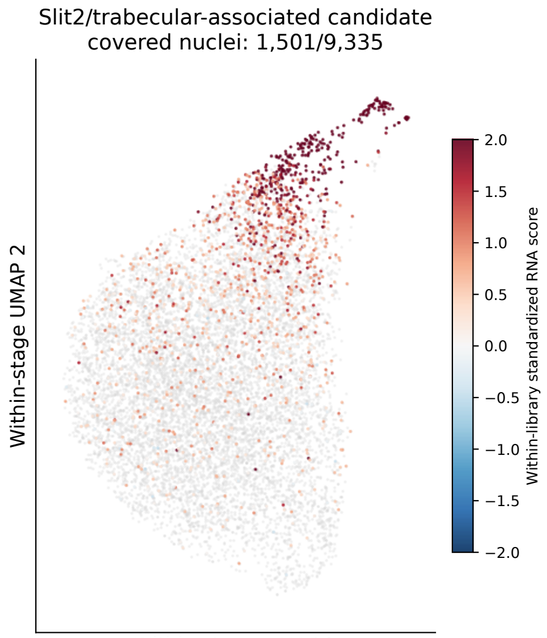
# P21: ventricular microenvironment-associated state candidates · n=9,335

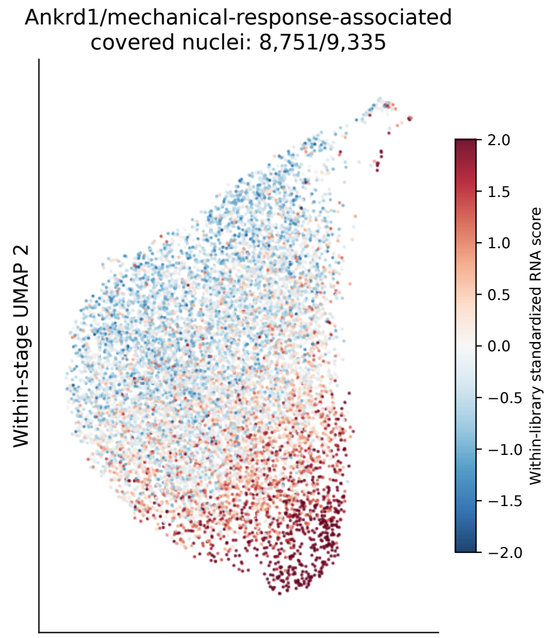


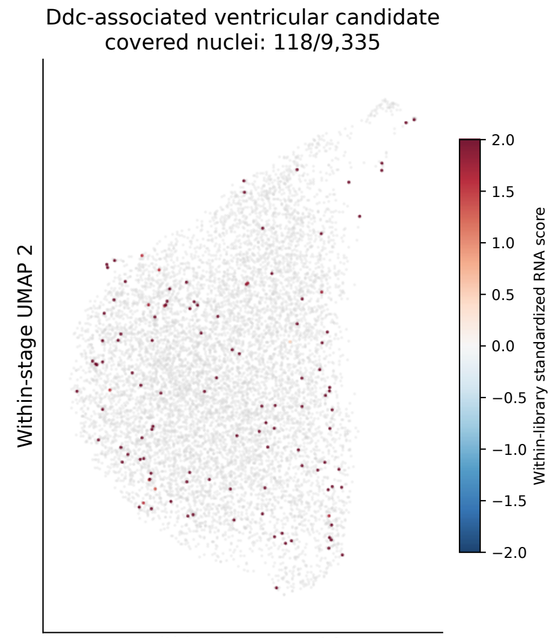

| Subcluster | Nuclei | Hypothesis-aligned label | Evidence / sensitivity status | Reference nuclei |
|---|---|---|---|---|
| P21_M0 | 1,778 | Marker-defined: Lrtm1/Stk39 | candidate/review | 0 |
| P21_M1 | 2,024 | Uncharacterized ventricular-enriched group | candidate/review | 0 |
| P21_M2 | 1,372 | Ankrd1/mechanical-response-associated | candidate/review | 0 |
| P21_M3 | 2,380 | Slit2/trabecular-associated candidate | supported RNA state | 56 |
| P21_M4 | 1,709 | Ankrd1/mechanical-response-associated | candidate/review | 0 |
| P21_M5 | 72 | Slit2/trabecular-associated candidate | candidate/review | 72 |

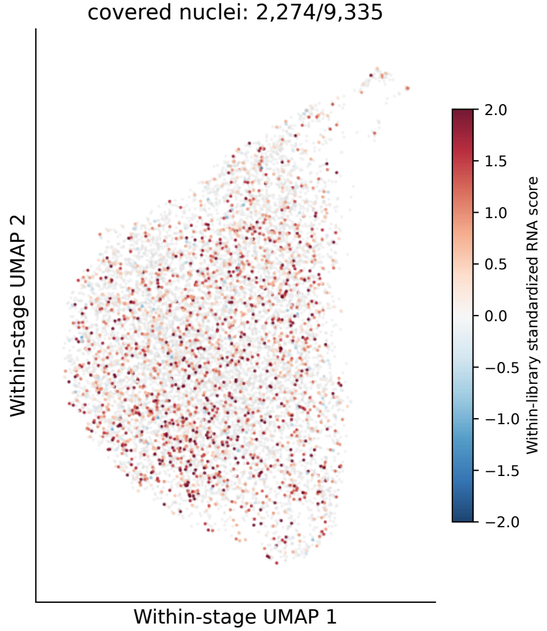
Same stage coordinates in all three panels · Ambiguous Slit2-overlap cells are a reference, not verified conduction or Purkinje cells
Labels are RNA-based hypotheses, not measured wall location, oxygen exposure or mechanical force. Other ages have independent coordinates.

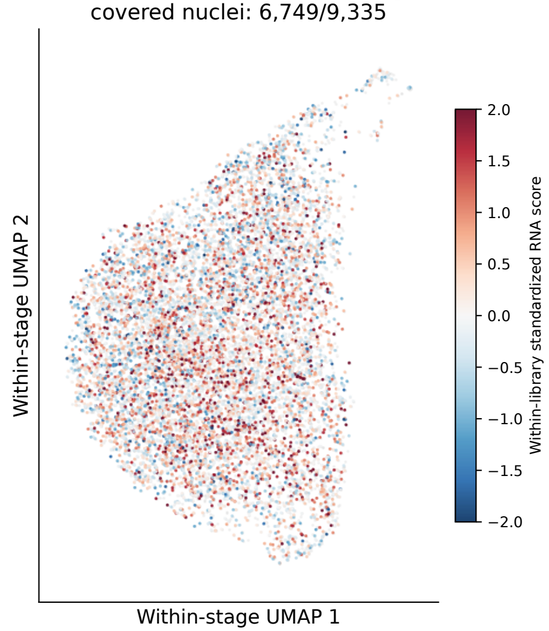
# P21: candidate spatial, oxygen-response and mechanical programs

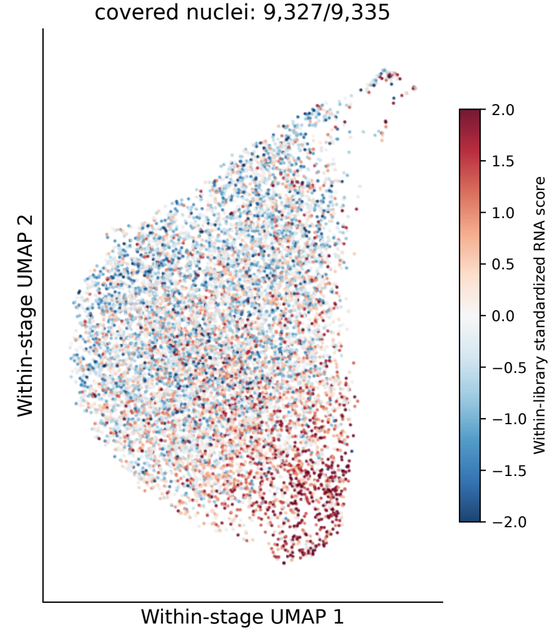



Same UMAP and axis limits throughout · Fixed color limits −2 to +2; scores are relative within libraries, not comparable oxygen measurements
Gray: nonfinite score or detected program-gene coverage <20% where coverage is supplied. Programs can overlap and form gradients.
RNA scores do not verify wall position, hypoxia, local force, or a discrete biological subtype.